

Spectral Structure of Mixed Negative-Type and Positive-Type Kernels

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1 Introduction

This paper studies a two-parameter family of compact, self-adjoint operators arising from mixtures of negative-type and positive-type kernels. The central object is the mixed kernel

$$K(a, b) = aK^{(L)} + bK^{(E)},$$

where $K^{(L)}$ is conditionally negative definite and $K^{(E)}$ is positive definite. The associated centered operator

$$D(a, b) = -\frac{1}{2}H(aK^{(L)} + bK^{(E)})H$$

acts on the mean-zero subspace $L_0^2(X)$ and exhibits nontrivial spectral behavior as (a, b) varies.

The purpose of this work is to analyze the sign structure of the eigenvalues of $D(a, b)$, the existence and uniqueness of critical curves where eigenvalues cross zero, and the stability of the positive direction induced by the negative-type component. The results are operator-theoretic and rely on classical tools such as the min-max principle, perturbation theory, and spectral continuity.

2 Kernel Background

Let X be a compact domain. A kernel $K : X \times X \rightarrow \mathbb{R}$ induces an integral operator

$$(T_K f)(x) = \int_X K(x, y) f(y) dy.$$

The centering operator H is defined by

$$(Hf)(x) = f(x) - \int_X f(y) dy,$$

and the centered kernel operator is

$$D_K = -\frac{1}{2}HKKH.$$

A kernel K is *positive definite* if

$$\sum_{i,j} c_i c_j K(x_i, x_j) \geq 0$$

for all choices of coefficients. A kernel is *conditionally negative definite* if the same inequality holds with $\sum_i c_i = 0$ and the sign reversed. These two classes produce operators with opposite spectral signatures: positive-type kernels yield negative semidefinite D_K , while negative-type kernels yield operators with exactly one positive eigenvalue.

3 Negative-Type and Positive-Type Components

Let $K^{(L)}$ be conditionally negative definite. Its centered operator

$$D^{(L)} = -\frac{1}{2}HK^{(L)}H$$

has exactly one positive eigenvalue $\lambda_1^{(L)} > 0$ and all others strictly negative. This follows from Schoenberg's characterization and the Krein–Rutman theorem applied to the cone of nonnegative functions.

Let $K^{(E)}$ be positive definite. Its centered operator

$$D^{(E)} = -\frac{1}{2}HK^{(E)}H$$

is negative semidefinite, with eigenvalues $\mu_k^{(E)} \leq 0$.

The mixed operator is

$$D(a, b) = aD^{(L)} + bD^{(E)}.$$

Since both components are compact and self-adjoint, $D(a, b)$ depends smoothly on (a, b) in operator norm, and each eigenvalue $\lambda_k(a, b)$ is a continuous function of (a, b) .

4 Mixed Operator and Main Structural Theorem

The operator $D(a, b)$ interpolates between a rank-one positive direction and a negative-semidefinite perturbation. The min–max principle implies that the top eigenvalue

$$\lambda_1(a, b) = \sup_{\|f\|=1} \langle f, D(a, b)f \rangle$$

is strictly increasing in a and strictly decreasing in b . This monotonicity is the key to understanding the structure of the mixed regime.

Theorem 1 (Persistence and Unique Crossing of the Positive Direction). *Let $D^{(L)}$ and $D^{(E)}$ be the centered operators associated to a negative-type and positive-type kernel respectively. Then:*

1. *There exists $\varepsilon > 0$ such that for all (a, b) with $a > 0$ and $|b| < \varepsilon$, the operator $D(a, b)$ has exactly one positive eigenvalue. Thus the positive direction induced by $D^{(L)}$ persists in an open neighborhood of the ray $(a, 0)$.*
2. *For each fixed $a > 0$, there exists a unique $b_c(a) > 0$ such that the top eigenvalue satisfies*

$$\lambda_1(a, b_c(a)) = 0,$$

with $\lambda_1(a, b) > 0$ for $b < b_c(a)$ and $\lambda_1(a, b) < 0$ for $b > b_c(a)$. The crossing is unique.

Proof Sketch. At $(a, 0)$, the operator is $aD^{(L)}$, which has exactly one positive eigenvalue. By continuity of eigenvalues, this persists for small b .

Fix $a > 0$. At $b = 0$, $\lambda_1(a, 0) = a\lambda_1^{(L)} > 0$. For sufficiently large b , the negative semidefinite operator $bD^{(E)}$ dominates, and the min-max principle implies $\lambda_1(a, b) < 0$. By continuity, there exists $b_c(a)$ with $\lambda_1(a, b_c(a)) = 0$.

Uniqueness follows from the strict monotonicity of $\lambda_1(a, b)$ in b , which is a consequence of the Rayleigh quotient and the negativity of $D^{(E)}$. $\square$

Corollary 1 (Uniqueness of the Positive-Direction Boundary). *For each fixed $a > 0$, the set of points (a, b) where $D(a, b)$ has exactly one positive eigenvalue is separated from the region where it has none by a single smooth curve $b = b_c(a)$. No additional positive directions can appear or disappear along this boundary.*

5 Phase Diagram

The mixed operator

$$D(a, b) = aD^{(L)} + bD^{(E)}$$

defines a two-parameter family of compact, self-adjoint operators on $L_0^2(X)$. The sign structure of its eigenvalues determines the qualitative behavior of the system. Theorem 4.1 establishes the existence and uniqueness of the boundary separating the region with one positive eigenvalue from the region with none.

5.1 5.1 Critical Curves

For each mode $k \geq 1$, define the critical set

$$\Gamma_k = \{(a, b) \in \mathbb{R}^2 : \lambda_k(a, b) = 0\}.$$

Since each eigenvalue $\lambda_k(a, b)$ is continuous in (a, b) , Γ_k is a closed subset of parameter space. When $\nabla \lambda_k(a, b) \neq 0$, the implicit function theorem implies that Γ_k is locally a smooth curve.

Theorem 4.1 implies that Γ_1 is a single smooth curve $b = b_c(a)$ separating the region with one positive eigenvalue from the region with none. Higher curves Γ_k may exist for $k \geq 2$, but they do not affect the uniqueness of the positive direction.

5.2 5.2 Lorentzian and Euclidean Regions

Define the signature index

$$\mathcal{I}(a, b) = p(a, b),$$

where $p(a, b)$ is the number of positive eigenvalues of $D(a, b)$. Theorem 4.1 implies that

$$\mathcal{I}(a, b) = \begin{cases} 1, & b < b_c(a), \\ 0, & b > b_c(a). \end{cases}$$

Thus the phase diagram contains two primary regions:

- the *positive-direction region* $\{(a, b) : b < b_c(a)\}$,
- the *negative-semidefinite region* $\{(a, b) : b > b_c(a)\}$.

The boundary between them is the unique curve Γ_1 .

5.3 5.3 Higher-Mode Transitions

For $k \geq 2$, the eigenvalues $\lambda_k(a, b)$ may cross zero at additional parameter values. These crossings produce higher critical curves Γ_k . Their existence depends on the decay rates of the spectra of $D^{(L)}$ and $D^{(E)}$. If the negative spectrum of $D^{(E)}$ decays slowly, multiple crossings may occur; if it decays rapidly, only finitely many crossings exist.

These higher-mode transitions do not affect the uniqueness of the positive direction but contribute to the structure of the spectral signature field described in Section 6.

6 Spectral Signature Field

The operator family $D(a, b)$ induces a spectral object that records the signs and magnitudes of all eigenvalues across parameter space. This structure provides a precise mathematical description of how the spectrum changes as (a, b) varies.

6.1 6.1 Definition

Define the spectral signature field

$$\mathcal{G}(a, b) = \{\lambda_k(a, b)\}_{k \geq 1},$$

where the eigenvalues are ordered by decreasing magnitude. The sign pattern of $\mathcal{G}(a, b)$ determines the signature

$$(p(a, b), n(a, b)),$$

with $p(a, b)$ the number of positive eigenvalues and $n(a, b)$ the number of negative ones.

6.2 6.2 Geometric Regions

The spectral signature field partitions parameter space into regions where the signature is constant. Theorem 4.1 implies that the region with one positive eigenvalue is separated from the region with none by a single smooth curve $b = b_c(a)$. Higher-mode transitions produce additional regions where the signature takes the form $(k, N - k)$ for $k \geq 2$.

These regions are determined entirely by the sign structure of the eigenvalues and do not rely on commutativity or special symmetry assumptions.

6.3 6.3 Spectral Invariants

The spectral signature field admits several invariants:

- the signature index $\mathcal{I}(a, b)$,
- the critical curves Γ_k ,
- the spectral flow along paths in parameter space,
- the persistence intervals $\mathcal{P}_k = \{(a, b) : \lambda_k(a, b) > 0\}$,
- asymptotic invariants determined by the decay rates of the spectra of $D^{(L)}$ and $D^{(E)}$.

These invariants determine the topology of the phase diagram and the structure of the spectral signature field.

6.4 Stability of the Positive Direction

Theorem 4.1 implies that the positive direction induced by $D^{(L)}$ is topologically stable: it persists in an open region and disappears exactly once along the boundary curve $b = b_c(a)$. No additional positive directions can appear or disappear along this boundary.

This stability is a structural property of the mixed operator and follows from the monotonicity of the top eigenvalue under negative-semidefinite perturbations.

7 Discussion of Spectral Behavior

The mixed operator

$$D(a, b) = aD^{(L)} + bD^{(E)}$$

exhibits spectral behavior governed by the interplay between a single positive direction and a negative-semidefinite perturbation. Theorem 4.1 establishes the structural stability of the positive direction and the uniqueness of its zero-crossing. Beyond the top eigenvalue, the remaining spectrum reflects the relative decay rates of the negative eigenvalues of $D^{(L)}$ and $D^{(E)}$.

7.1 Mode-by-Mode Behavior

For each mode $k \geq 2$, the eigenvalue $\lambda_k(a, b)$ satisfies

$$\lambda_k(a, b) = a\lambda_k^{(L)} + b\mu_k^{(E)}.$$

Since both $\lambda_k^{(L)}$ and $\mu_k^{(E)}$ are negative, the sign of $\lambda_k(a, b)$ is determined by the relative magnitudes of the two terms. Modes for which $|\lambda_k^{(L)}|$ decays slowly compared to $|\mu_k^{(E)}|$ may exhibit zero-crossings, producing higher critical curves Γ_k .

7.2 Asymptotic Regimes

If $D^{(L)}$ arises from a fractional kernel $|x - y|^\alpha$ on a d -dimensional domain, its spectrum satisfies

$$\lambda_n^{(L)} \sim -c_{\alpha, d} n^{-(2+\alpha)/d}.$$

If $D^{(E)}$ arises from a Laplacian or Matérn kernel, its spectrum typically satisfies

$$\mu_n^{(E)} \sim -C n^{-\beta}$$

for some $\beta > 0$. The number of higher critical curves is determined by the comparison of these decay rates. If $(2 + \alpha)/d < \beta$, only finitely many crossings occur; if $(2 + \alpha)/d > \beta$, infinitely many crossings may occur.

8 Spectral Invariants

The spectral signature field $\mathcal{G}(a, b)$ admits several invariants that characterize the structure of the mixed operator. These invariants are operator-theoretic and do not rely on geometric interpretation.

8.1 8.1 Signature Index

The signature index

$$\mathcal{I}(a, b) = p(a, b)$$

is the number of positive eigenvalues of $D(a, b)$. Theorem 4.1 implies that $\mathcal{I}(a, b)$ changes only along the unique curve Γ_1 .

8.2 8.2 Critical Curves

For each mode k , the critical curve

$$\Gamma_k = \{(a, b) : \lambda_k(a, b) = 0\}$$

is a spectral invariant. The ordering of these curves determines the topology of the spectral signature field.

8.3 8.3 Spectral Flow

For any path $\gamma : [0, 1] \rightarrow \mathbb{R}^2$, the spectral flow

$$\text{SF}(\gamma)$$

counts the net number of eigenvalues crossing zero. The uniqueness of the positive-direction crossing implies that the spectral flow of the top eigenvalue is always ± 1 along any path connecting the positive-direction region to the negative-semidefinite region.

8.4 8.4 Persistence Intervals

For each mode k , the persistence interval

$$\mathcal{P}_k = \{(a, b) : \lambda_k(a, b) > 0\}$$

is an invariant describing where the mode contributes a positive direction. Only $\mathcal{P}_1$ is nonempty for small b , and its boundary is the unique curve Γ_1 .

8.5 Asymptotic Invariants

The decay rates of $\lambda_n^{(L)}$ and $\mu_n^{(E)}$ determine the number and structure of higher critical curves. These asymptotic invariants govern the global behavior of the spectral signature field.

9 Applications and Interpretation

Although the mixed operator is defined abstractly, its spectral behavior has applications in settings where kernels encode relational structure.

9.1 Multi-Scale Structure Detection

Negative-type kernels emphasize large-scale separation, while positive-type kernels emphasize local similarity. The mixed operator interpolates between these regimes. The sign of the top eigenvalue indicates whether global structure or local similarity dominates.

9.2 Diffusion and Ultrametric Transitions

Gaussian kernels generate diffusion-like behavior, while fractional kernels generate ultrametric-like behavior. The mixed operator captures transitions between these behaviors. The unique critical curve Γ_1 marks the point where diffusion dominates sufficiently to eliminate the positive direction.

9.3 Graph and Network Operators

On graphs, kernels such as commute-time, resistance distance, and diffusion kernels can be mixed. The spectral signature field describes how global and local connectivity interact. The existence of a unique positive-direction boundary persists in this discrete setting.

9.4 Kernel Mixtures in Machine Learning

Kernel mixtures are common in kernel learning. The mixed operator provides a spectral method for understanding how kernel weights affect the sign structure of the induced operator. The unique boundary curve $b = b_c(a)$ identifies the transition between regimes dominated by global versus local similarity.

10 Examples

We illustrate the mixed-operator framework with several examples. Each example demonstrates a distinct spectral behavior.

10.1 10.1 Fractional Distance Plus Gaussian Kernel

Let $X = [0, 1]$ and consider

$$K^{(L)}(x, y) = |x - y|^\alpha, \quad K^{(E)}(x, y) = e^{-\gamma|x-y|^2}.$$

The mixed operator exhibits a single critical curve Γ_1 , separating the region with one positive eigenvalue from the region with none. Higher-mode crossings do not occur due to the rapid decay of the Gaussian spectrum.

10.2 10.2 Fractional Distance Plus Inner-Product Kernel

Let

$$K^{(E)}(x, y) = xy.$$

Since $D^{(E)}$ has rank one, only the top eigenvalue can cross zero. Thus the phase diagram contains only the unique curve Γ_1 .

10.3 10.3 Gaussian Plus Gaussian

If both kernels are positive definite, then $D(a, b)$ is negative semidefinite for all (a, b) . No positive direction exists, and no critical curves occur.

10.4 10.4 Fractional Distance Plus Laplacian Kernel

Let

$$K^{(E)}(x, y) = \min(x, y).$$

The Laplacian spectrum decays polynomially, producing multiple higher-mode crossings. The phase diagram contains the unique curve Γ_1 and additional curves Γ_k for $k \geq 2$.

10.5 10.5 Summary

These examples demonstrate:

- the uniqueness of the positive-direction boundary,
- the dependence of higher-mode crossings on spectral decay rates,
- the robustness of the mixed-operator framework across kernels.

Appendix A: Proof Sketches

This appendix provides strengthened proof sketches for several results stated in the main text. Each argument uses standard tools from spectral theory, perturbation theory, and harmonic analysis. Full proofs follow from established results in functional analysis.

A.1 Conditional Negative Definiteness of $|x - y|^\alpha$

For $0 < \alpha < 2$, Schoenberg's theorem states that a function φ on $\mathbb{R}^d$ is conditionally negative definite if and only if

$$\varphi(x) = c\|x\|^2 + \int_{\mathbb{R}^d} (1 - \cos(x \cdot t)) \nu(dt)$$

for some finite positive measure ν . The fractional power satisfies

$$|x|^\alpha = C_\alpha \int_0^\infty (1 - \cos(tx)) t^{-1-\alpha} dt,$$

which is of Schoenberg form. Thus for any coefficients with $\sum_i c_i = 0$,

$$\sum_{i,j} c_i c_j |x_i - x_j|^\alpha \leq 0.$$

This implies that the centered operator $D^{(L)}$ has exactly one positive eigenvalue.

A.2 Positivity of the Top Eigenfunction

Since $K^{(L)}$ is strictly positive off the diagonal, the operator $HK^{(L)}H$ is compact and strongly positive on the cone of nonnegative functions in $L_0^2(X)$. By the Krein–Rutman theorem, the top eigenvalue of $D^{(L)}$ is simple and its eigenfunction may be chosen strictly positive.

A.3 Fractional Asymptotics with Error Bounds

Let $f_n(x) = \cos(n\pi x)$ on $[0, 1]$. A stationary phase expansion of

$$Q_\alpha(f_n) = \iint f_n(x) f_n(y) |x - y|^\alpha dx dy$$

yields

$$Q_\alpha(f_n) = -c_\alpha n^{-(1+\alpha)} + O(n^{-(2+\alpha)}).$$

The error term follows from the standard one-dimensional stationary phase estimate applied to the oscillatory integral representation of $|x - y|^\alpha$. Since eigenvalues interlace with Rayleigh quotients, the same asymptotic holds for $\lambda_n^{(L)}$.

A.4 Negative Semidefiniteness of Centered Positive-Type Kernels

If $K^{(E)}$ is positive definite, then for any $f \in L^2(X)$,

$$\iint f(x)f(y)K^{(E)}(x,y)dx dy \geq 0.$$

For $f \in L_0^2(X)$, write $f = Hg$. Then

$$(f, D^{(E)}f) = -\frac{1}{2}(Hg, K^{(E)}Hg) \leq 0.$$

Thus $D^{(E)}$ is negative semidefinite.

A.5 Continuity of Eigenvalues

Since $D(a, b) = aD^{(L)} + bD^{(E)}$ is a linear combination of compact self-adjoint operators, it depends continuously on (a, b) in operator norm. By classical perturbation theory, each eigenvalue $\lambda_k(a, b)$ is a continuous function of (a, b) .

A.6 Existence of Critical Curves

Let $\lambda_k(a, b)$ be the k -th eigenvalue of $D(a, b)$. Since $\lambda_k(a, 0) = a\lambda_k^{(L)}$ and $\lambda_k(0, b) = b\mu_k^{(E)}$, the intermediate value theorem guarantees that if $\lambda_k^{(L)} > 0$ and $\mu_k^{(E)} < 0$, then there exists (a, b) with $\lambda_k(a, b) = 0$. The set of solutions forms a smooth curve by the implicit function theorem, provided $\nabla \lambda_k(a, b) \neq 0$ at the crossing.

A.7 Higher-Dimensional Fractional Asymptotics

On $X \subset \mathbb{R}^d$, the kernel $|x - y|^\alpha$ remains conditionally negative definite. The spectral asymptotics follow from the local singularity of $|x - y|^\alpha$ near the diagonal and the density of high-frequency modes in dimension d , yielding

$$\lambda_n^{(L)} \sim -c_{\alpha, d} n^{-(2+\alpha)/d}.$$

Appendix B: Non-Commuting Perturbation Theory

This appendix strengthens the analysis of the mixed operator in the non-commuting case. No commutativity assumptions are required for the existence or uniqueness of the positive-direction boundary.

B.1 Eigenvalue Derivative Formula

Let $D(a, b) = aD^{(L)} + bD^{(E)}$ and let $\lambda_k(a, b)$ be a simple eigenvalue with normalized eigenfunction $v_k(a, b)$. Kato's perturbation theory implies the derivative formula

$$\frac{\partial}{\partial b} \lambda_k(a, b) = \langle v_k(a, b), D^{(E)} v_k(a, b) \rangle.$$

Since $D^{(E)}$ is negative semidefinite, this derivative is nonpositive for all (a, b) . For the top eigenvalue,

$$\frac{\partial}{\partial b} \lambda_1(a, b) < 0,$$

because $v_1(a, b)$ is strictly positive and $D^{(E)}$ is strictly negative on nonconstant functions. This strict monotonicity ensures the uniqueness of the zero crossing.

B.2 Resolvent Identity

Let $R(a, b; z) = (D(a, b) - zI)^{-1}$ be the resolvent. The resolvent identity

$$R(a, b; z) - R(a, b'; z) = (b' - b)R(a, b; z)D^{(E)}R(a, b'; z)$$

provides control over the movement of eigenvalues under changes in b . Since $D^{(E)}$ is negative semidefinite, the resolvent decreases monotonically in b in the sense of quadratic forms.

B.3 Monotonicity and Uniqueness of the Crossing

Combining the derivative formula and the resolvent identity yields:

- $\lambda_1(a, b)$ is strictly decreasing in b ,
- $\lambda_k(a, b)$ is nonincreasing for all k ,
- no eigenvalue can cross zero more than once.

Thus the positive-direction boundary curve $b = b_c(a)$ is unique even when $D^{(L)}$ and $D^{(E)}$ do not commute.

B.4 Higher-Mode Behavior

For $k \geq 2$, the derivative formula implies that $\lambda_k(a, b)$ is nonincreasing in b . Zero-crossings occur only when the negative spectrum of $D^{(L)}$ decays slowly relative to that of $D^{(E)}$. The number of crossings is determined by the asymptotic comparison of the spectra.

B.5 Summary

The non-commuting perturbation theory shows:

- the positive direction is structurally stable,
- the boundary curve $b = b_c(a)$ is unique,
- higher-mode crossings depend on spectral decay rates,
- commutativity is not required for any structural result.